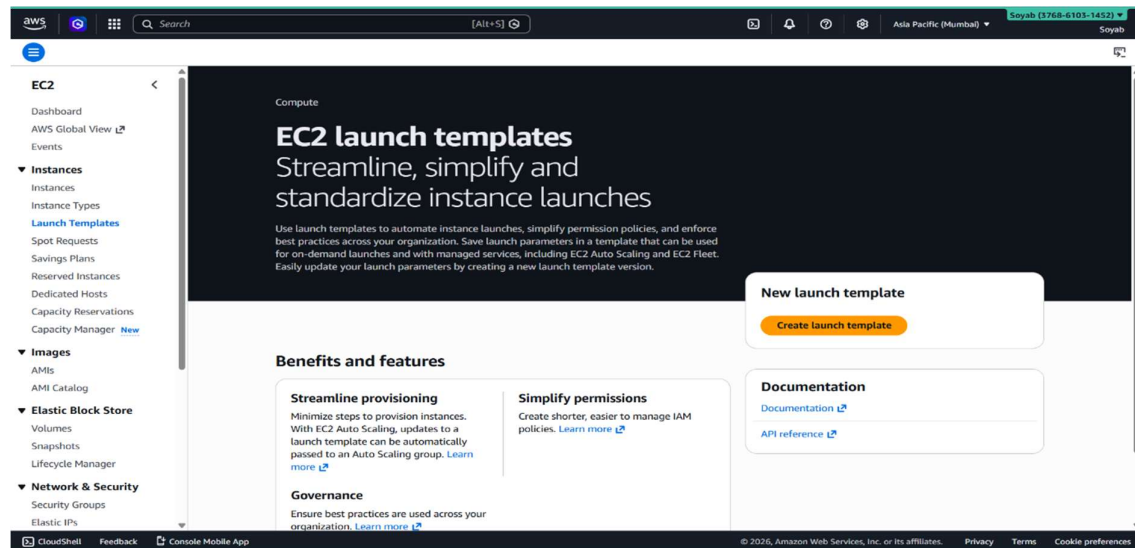


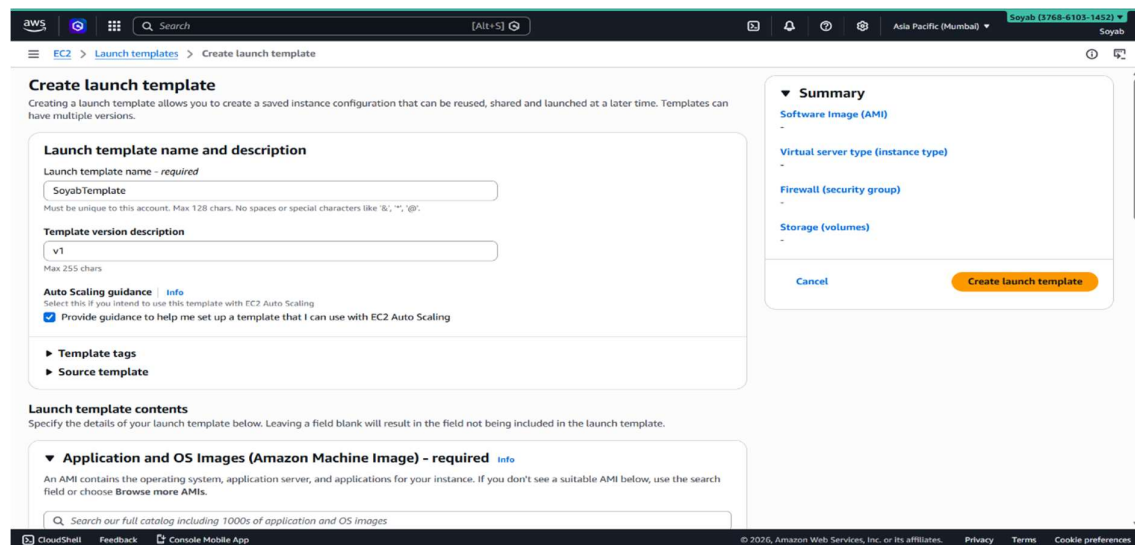
Assignment : 11

Build scaling plans in AWS that balance the load on different EC2 instances.

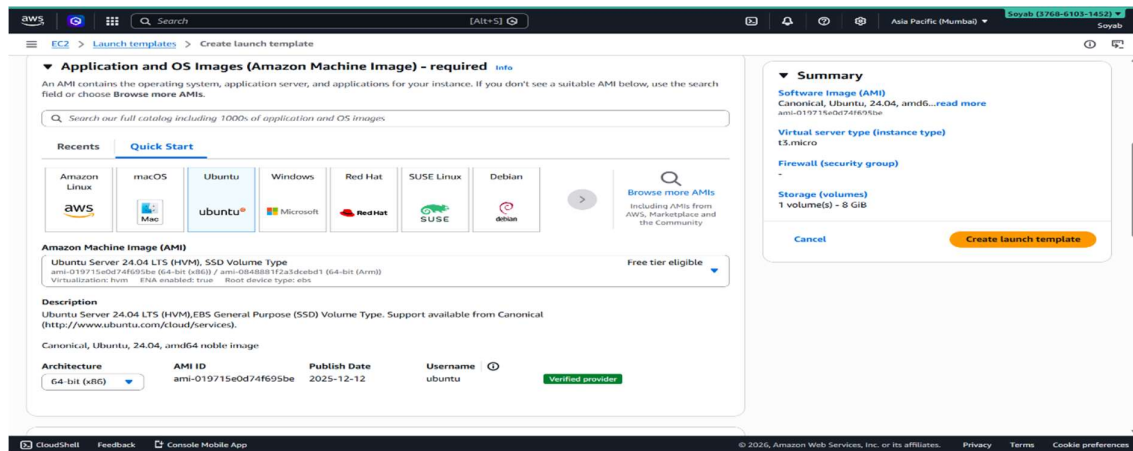
Step 1: From the lefthand side menu bar, go to Launch Templates and click on “Launch Template”. Name the Template and set the version as “v1”, also check the Auto scaling guidance box.



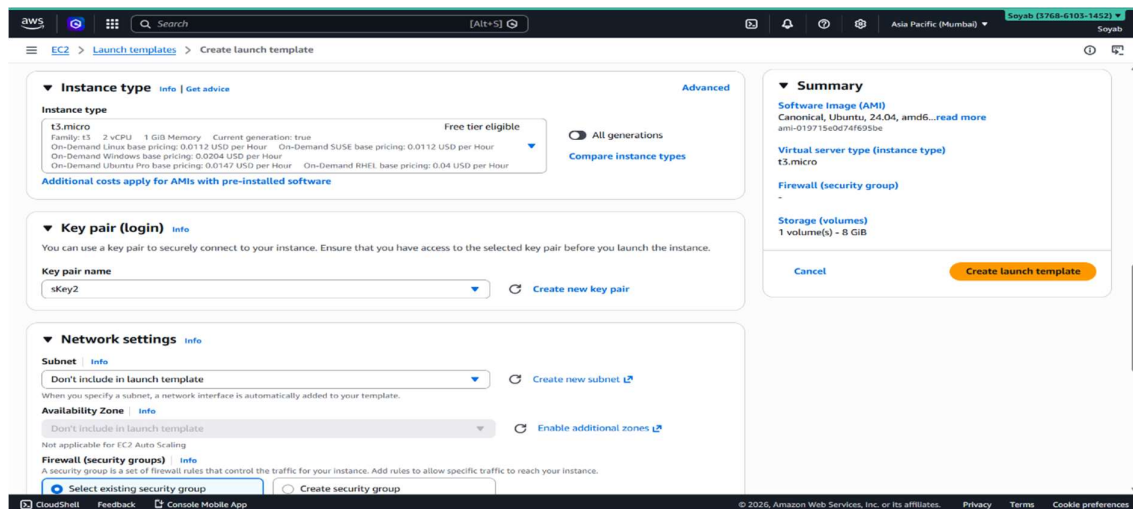
Step 2: Name the Template and set the version as “v1”, also check the Auto scaling guidance box.



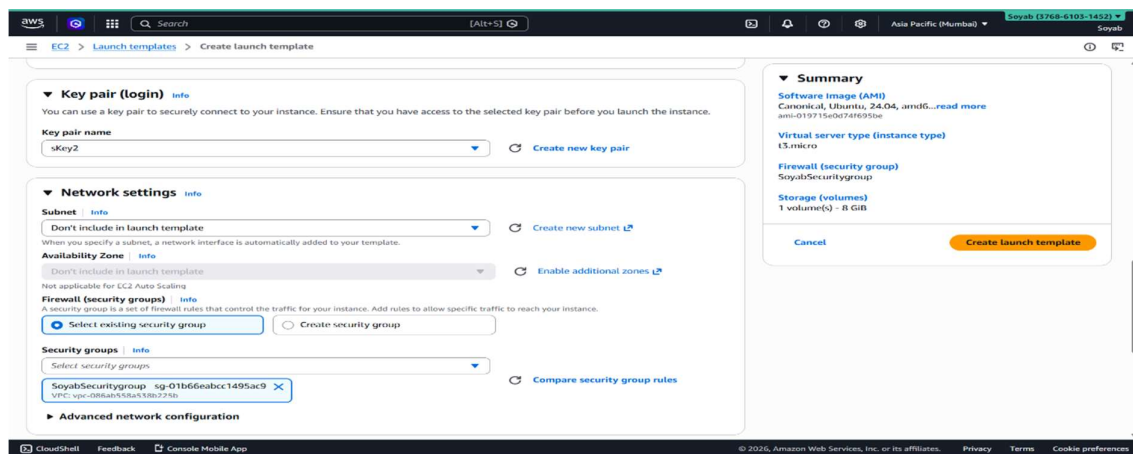
Step 3: Scroll down to Application and OS > Quick Start > Ubuntu.



Step 4: In the instance type, select t3.micro as it is the Free tier eligible instance.



Step 5: Network Settings > Firewall (security groups) > Select Existing groups. Now select the above made security group.



Step 6: In the Advanced details, write the below mentioned code,

#!/bin/bash

apt-get update

apt-get upgrade

apt-get install-y nginx

systemctl start nginx

systemctl enable nginx

apt-get install-y git

curl-sL https://deb.nodesource.com/setup_18.x | sudo-E bash -

apt-get install-y nodejs

git clone https://github.com/TheWolf313 /Web_testing.git

cd Web_testing

npm install

node index.js

And theclick on “Create Launch Template”.

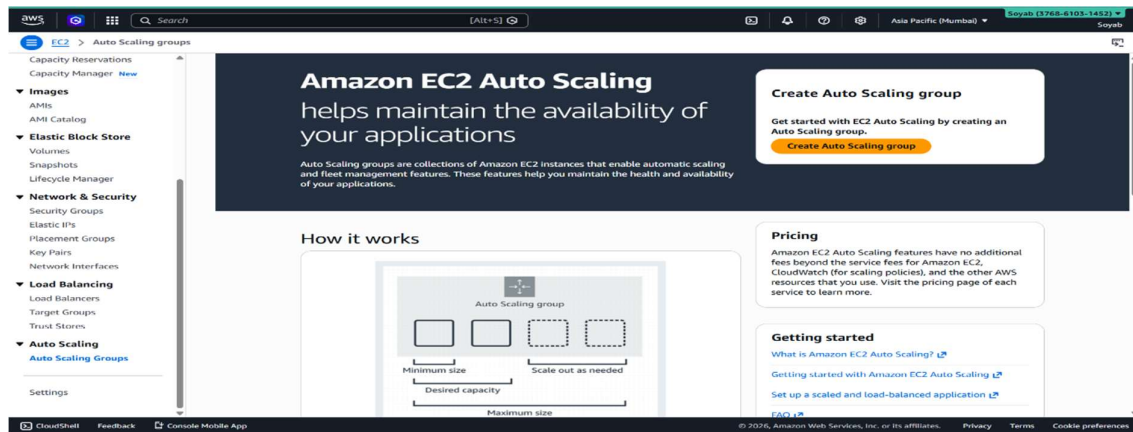
The screenshot shows the AWS Management Console interface for creating a launch template. The top navigation bar includes the AWS logo, search, and navigation icons. The main content area is titled 'Create launch template' and is divided into two panels. The left panel, 'User data - optional', contains a text area with a script for setting up a Linux instance. The script includes commands for updating the system, installing nginx and git, cloning a repository, and running a Node.js application. The right panel, 'Summary', displays the configuration details for the launch template, including the software image (AMI), virtual server type (instance type), firewall (security group), and storage (volumes). A 'Create launch template' button is located at the bottom right of the summary panel. Below the main form, there is a 'Success' message indicating the template was created, followed by 'Next Steps' and 'Actions log'.

Success
Successfully created SoyabTemplate(1l-0G83fc0b736b-1f1d4).

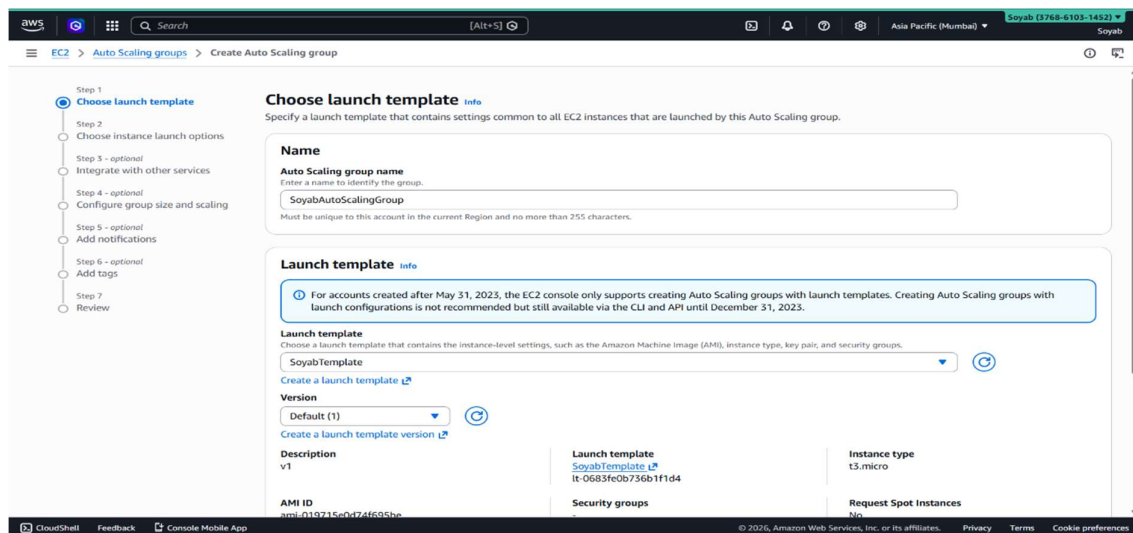
Next Steps
Launch an instance
With On-Demand Instances, you pay for compute capacity by the second (for Linux, with a minimum of 60 seconds) or by the hour (for all other operating systems) with no long-term commitments or upfront payments. Launch an On-Demand Instance from your launch template.
Launch Instance from this template
Create an Auto Scaling group from your template
Amazon EC2 Auto Scaling helps you maintain application availability and allows you to scale your Amazon EC2 capacity up or down automatically according to conditions you define. You can use Auto Scaling to help ensure that you are running your desired number of Amazon EC2 instances during demand spikes to maintain performance and decrease capacity during lulls to reduce costs.
Create Auto Scaling group
Create Spot Fleet
A Spot Instance is an unused EC2 instance that is available for less than the On-Demand price. Because Spot Instances enable you to request unused EC2 instances at steep discounts, you can lower your Amazon EC2 costs significantly. The hourly price for a Spot Instance (of each instance type in each Availability Zone) is set by Amazon EC2, and adjusted gradually based on the long-term supply of and demand for Spot instances. Spot instances are well suited for data analysis, batch jobs, background processing, and optional tasks.
Create Spot Fleet

[View launch templates](#)

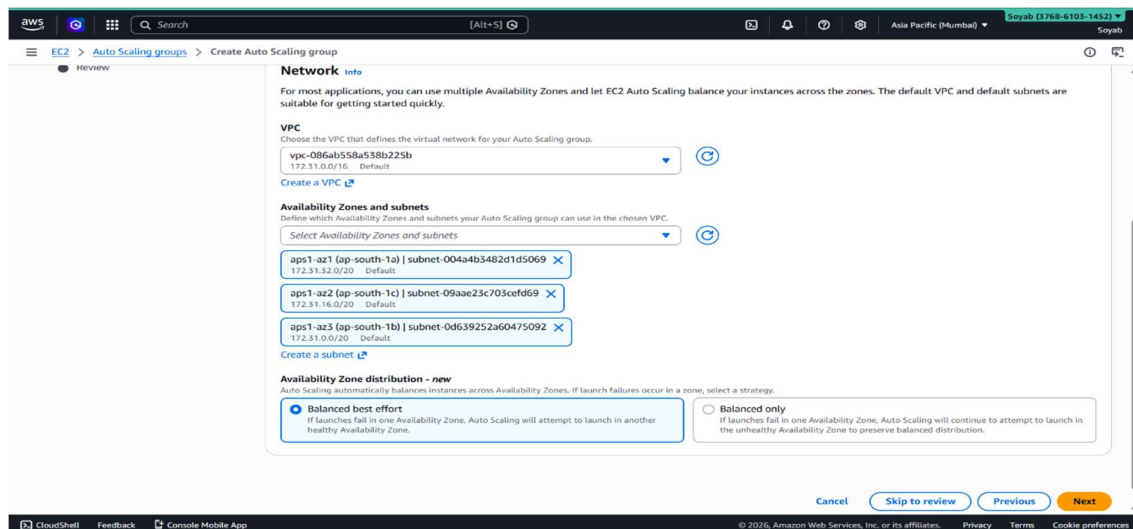
Step 7: Go to Auto Scaling Groups in the lefthand side menu.



Step 8: Click on “Create Auto Scaling Group”, name the group, select the above made template in the LaunchTemplate section, and select the version as default(1).



Step 9: In step 2, select all 3 Availability Zones and Subnets and click on next.



Step 10: In step 3, select the below shown options and Create a new target group in Default Routing and click on Next.

The screenshot displays the AWS Management Console interface for creating an Auto Scaling group. The breadcrumb navigation shows 'EC2 > Auto Scaling groups > Create Auto Scaling group'. A progress bar on the left indicates the current step is 'Integrate with other services - optional'.

Integrate with other services - optional
Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

- ☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.
- ☐ Attach to an existing load balancer
Choose from your existing load balancers.
- ☒ Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to a new load balancer

Load balancer type
Choose from the load balancer types offered below. Type selection cannot be changed after the load balancer is created. If you need a different type of load balancer than those offered here, visit the [Load Balancing console](#).

- ☒ Application Load Balancer
HTTP, HTTPS
- ☐ Network Load Balancer
TCP, UDP, TLS

Load balancer name
Name cannot be changed after the load balancer is created.
SoyabAutoScalingGroup-1

Load balancer scheme
Scheme cannot be changed after the load balancer is created.

- ☐ Internal
- ☒ Internet-facing

Network mapping
Your new load balancer will be created using the same VPC and Availability Zone selections as your Auto Scaling group. You can select different subnets and add subnets from additional Availability Zones.

VPC
vpc-086ab558a538b225b

Availability Zones and subnets
You must select a single subnet for each Availability Zone enabled. Only public subnets are available for selection to support DNS resolution.

- ☒ ap-south-1a
Select a subnet: subnet-004a4b3482d1d5069
- ☒ ap-south-1b
Select a subnet: subnet-0d639252a60475092
- ☒ ap-south-1c
Select a subnet: subnet-09aae23c705cfd69

Listeners and routing
If you require secure listeners, or multiple listeners, you can configure them from the [Load Balancing console](#) after your load balancer is created.

Protocol
HTTP

Port
4000

Default routing (forward to)
Create a target group

New target group name
An instance target group with default settings will be created.
SoyabAutoScalingGroup-1

Tags - optional
Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add tag](#)
\$0 remaining

Step 12: Select Desired capacity as 2, Min desired capacity as 2 and Max desired capacity as 3.

Configure group size and scaling - optional [Info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

2

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity **Max desired capacity**

2 3

Equal or less than desired capacity Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Step 13: In Automatic Scaling, Choose Target tracking scaling policy, set the policy name, Target value as 20 and Instance Warmup as 300 seconds and click on Next.

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

Target Tracking Policy

Metric type [Info](#)

Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization

Target value

20

Instance warmup [Info](#)

300 seconds

☐ Disable scale in to create only a scale-out policy

Instance maintenance policy [Info](#)

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

☒ Mixed behavior

For rebalancing events, new instances will launch before terminating others.

☐ Prioritize availability

Launch new instances and wait for them to be ready before terminating

☐ Control costs

Terminate and launch instances at the same time. This allows you to go below

☐ Flexible

Custom behavior

Set custom values for the minimum and maximum amount of available

Step 14: In step 5 & 6, click on Next. In step 7, after reviewing the wwhole Auto Scaling Group, click on Create Auto Scaling Group.

Add notifications - optional [Info](#)

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

[Add notification](#)

[Cancel](#) [Skip to review](#) [Previous](#) [Next](#)

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Asia Pacific (Mumbai)

Soyab (3768-6103-1452)

Soyab

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1: Choose launch template

Step 2: Choose instance launch options

Step 3 - optional: Integrate with other services

Step 4 - optional: Configure group size and scaling

Step 5 - optional: Add notifications

Step 6 - optional: Add tags

Step 7: Review

Add tags - optional info

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group.

Tags (0)

No tags

Add tag

50 remaining

Cancel

Previous

Next

CloudShell

Feedback

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Step 5 - optional: Add notifications

Step 6 - optional: Add tags

Step 7: Review

Review info

Step 1: Choose launch template

Group details

Auto Scaling group name
SoyabAutoScalingGroup

Launch template

Launch template
SoyabTemplate [↗](#)

Version
Default

Description
v1

Step 2: Choose instance launch options

Network

VPC
vpc-086ab558a538b225b [↗](#)

Availability Zones and subnets

Availability Zone	Subnet	Subnet CIDR range
aps1-az1 (ap-south-1a)	subnet-004a4b3482d1d5069 ↗	172.31.32.0/20
aps1-az2 (ap-south-1c)	subnet-09aae23c703cfd69 ↗	172.31.16.0/20
aps1-az3 (ap-south-1b)	subnet-0d639252a60475092 ↗	172.31.0.0/20

CloudShell

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Asia Pacific (Mumbai)

Soyab (3768-6103-1452)

Soyab

EC2 > Auto Scaling groups

Auto Scaling groups (1) info

Search your Auto Scaling groups

Launch configurations

Launch templates [↗](#)

Actions [⌵](#)

Create Auto Scaling group

Name	Launch template/configuration ↗	Instances	Status	Desired capacity	Min	Max	Availability Zones	
SoyabAutoScalingGroup	SoyabTemplate Version Default	0	Updating capacity...	2	2	3	Availability Zones	Th

0 Auto Scaling groups selected

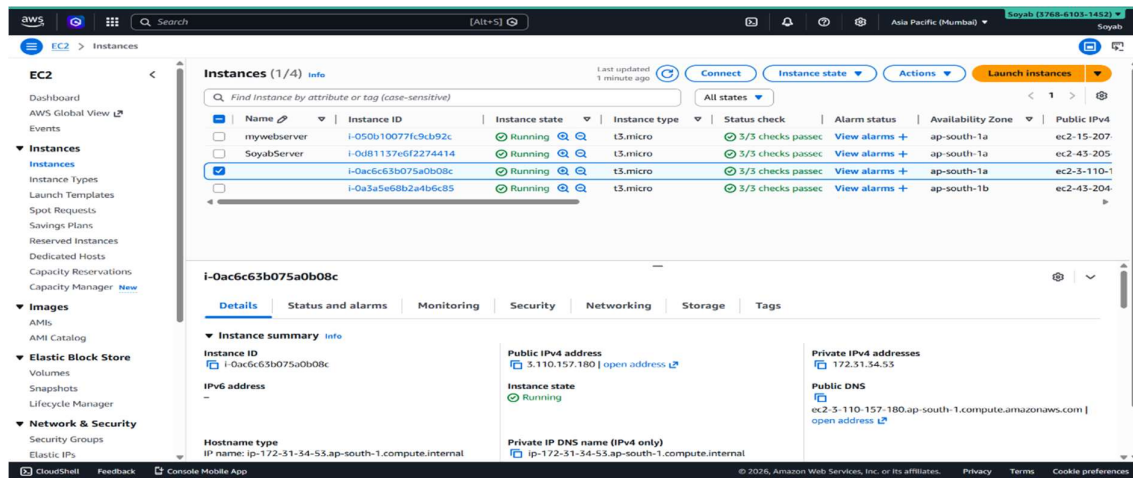
CloudShell

Feedback

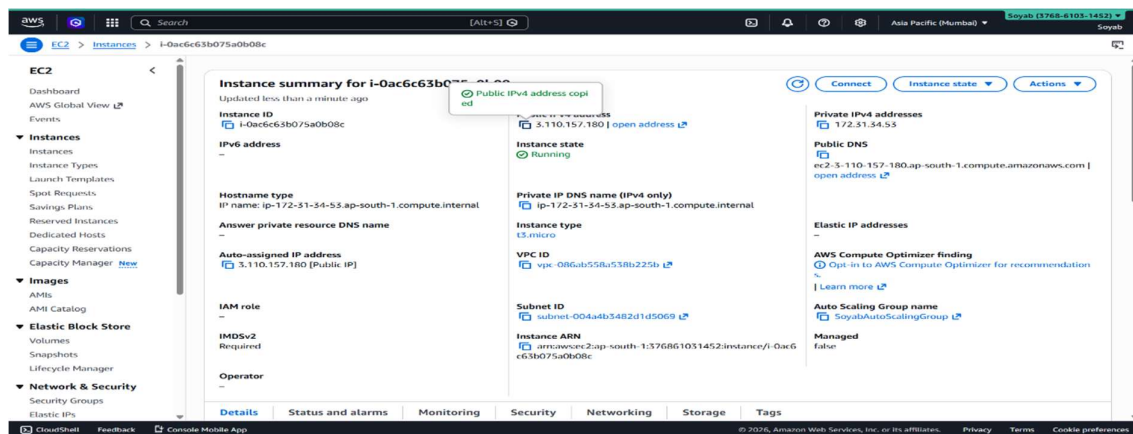
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Step 15: Go to Instances.

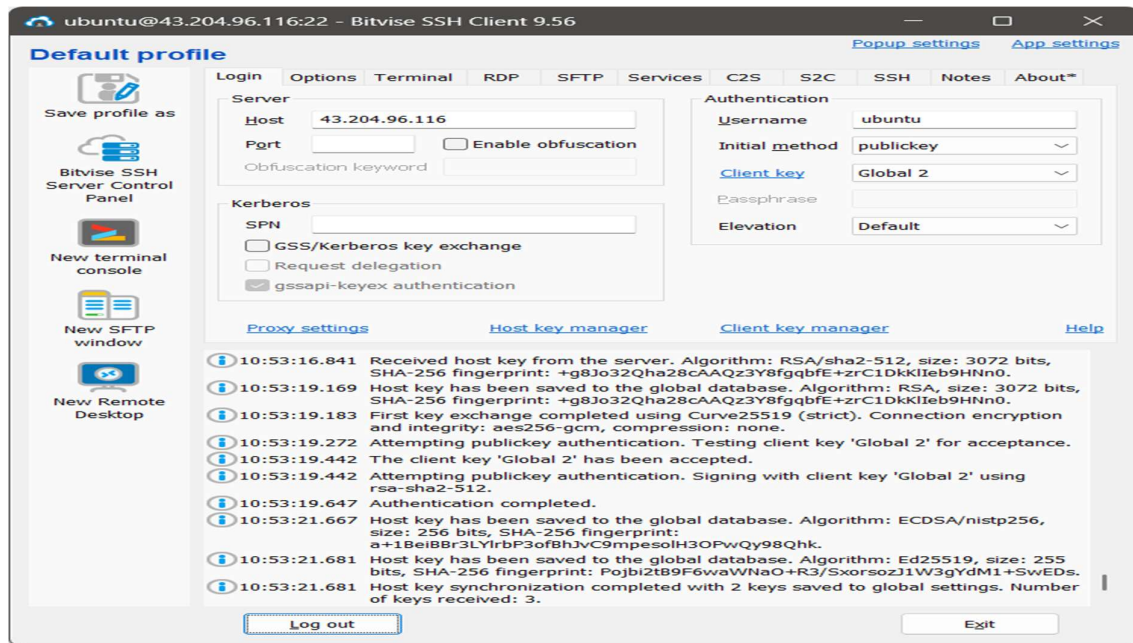


Step 16: Click on the first instance and copy its IPv4 address and run it in a separate incognito window to check if it's functional or not.



Hello MCKV

Step 17 : Copy the same IPv4 address in Bitwise SSH Client, import the newly made key pair and Log in.



Step 18: Open a new Terminal. Step 22: Create a new file using “nano” command and write the following block of code,#!/bin/bash

while true

do

echo "Running"

done.



Step 19: Run the file using the command.

Step 22: Maximize the CPU Utilization window to have a better view.

